

Thermal Properties of Matter

In our everyday experience, we constantly encounter the effects of heat and temperature. When we hold a hot cup of tea, we feel warmth flowing into our hands. When we leave ice cream out on a summer day, it melts. When we heat water in a kettle, it eventually boils and turns to steam. All these phenomena are manifestations of the thermal properties of matter — the ways in which substances respond to changes in temperature and the exchange of thermal energy. Understanding these properties is fundamental to physics and has enormous practical significance in engineering, meteorology, biology, and countless other fields.

This chapter provides a comprehensive exploration of how matter interacts with heat. We begin by carefully distinguishing between the concepts of heat and temperature — two terms that are often confused in everyday language but have very distinct physical meanings. We then examine how temperature is measured using various types of thermometers and temperature scales, and how the ideal gas equation connects the macroscopic properties of gases. The phenomenon of thermal expansion, the remarkable anomalous behaviour of water, the concepts of specific heat capacity and calorimetry, changes of state and latent heat, the three fundamental modes of heat transfer, and Newton's Law of Cooling are all explored with full mathematical treatment, physical reasoning, and practical applications.

The study of thermal properties underpins technologies ranging from refrigeration and air conditioning to thermal power generation, from the design of buildings and bridges to the manufacturing of precision instruments. By mastering these concepts, you will gain insight into phenomena as diverse as why lakes freeze from the top down, why a thermos flask keeps beverages hot, and why metals feel colder to the touch than wood at the same temperature.

10.1 Introduction

The sensation of hot and cold is one of the most fundamental human experiences. In physics, our understanding of thermal phenomena rests on two foundational concepts: heat and temperature. While these terms are used interchangeably in everyday language, they represent fundamentally different physical quantities with distinct definitions, units, and physical significance.

Temperature is a property of a body that determines the direction in which heat will flow when the body is placed in thermal contact with another body. It is a measure of the average kinetic energy of the molecules of a substance. When we say a body is "hot," we mean its molecules are moving rapidly; when it is "cold," its molecules are relatively sluggish. Temperature is what a thermometer measures.

Heat, on the other hand, is not a property of a body but rather energy in transit. It is the energy that flows between two bodies because of a temperature difference between them. Heat always flows spontaneously from a region of higher temperature to a region of lower temperature. Once heat has been transferred, it becomes part of the internal energy of the receiving body and is no longer called "heat." A body does not "contain" heat; rather, heat is the energy exchanged during a thermal interaction.

Heat vs Temperature: A Detailed Comparison

To appreciate the difference, consider this analogy: imagine two water tanks at different heights connected by a pipe. Water flows from the higher tank to the lower tank due to the height difference. Here, height corresponds to temperature, and the amount of water that flows corresponds to heat. The height determines the direction of flow, while the amount that flows depends on the height difference and the capacity of the tanks.

DEFINITION

Heat: The form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference. It flows spontaneously from higher to lower temperature. SI unit: Joule (J). Also measured in calories (1 cal = 4.186 J).

DEFINITION

Temperature: A physical quantity that expresses the degree of hotness or coldness of a substance. It is a measure of the average translational kinetic energy of the molecules. It determines the direction of spontaneous heat flow. SI unit: Kelvin (K).

HEAT VS TEMPERATURE

Heat is energy in transit (measured in Joules). **Temperature** is a property of the body that tells how hot or cold it is (measured in °C, °F, or K).

Analogy: Temperature is like height of water in a tank; heat is like the amount of water. A small cup of boiling water has high temperature but less heat than a bathtub of warm water.

Property	Heat	Temperature
What is it?	Form of energy	Degree of hotness
SI Unit	Joule (J)	Kelvin (K)
Depends on	Mass, specific heat, temp change	Average KE of molecules
Measured by	Calorimeter	Thermometer
Transfer	Flows from hot to cold	Does not flow

Key Points

- ✓ Heat is energy; temperature is a condition of the body
- ✓ Heat flows due to temperature difference (not due to heat difference)
- ✓ Two bodies at same temperature → no heat flow (thermal equilibrium)
- ✓ Zeroth Law: If A is in thermal equilibrium with B, and B with C, then A is in equilibrium with C

10.2 Temperature and Heat

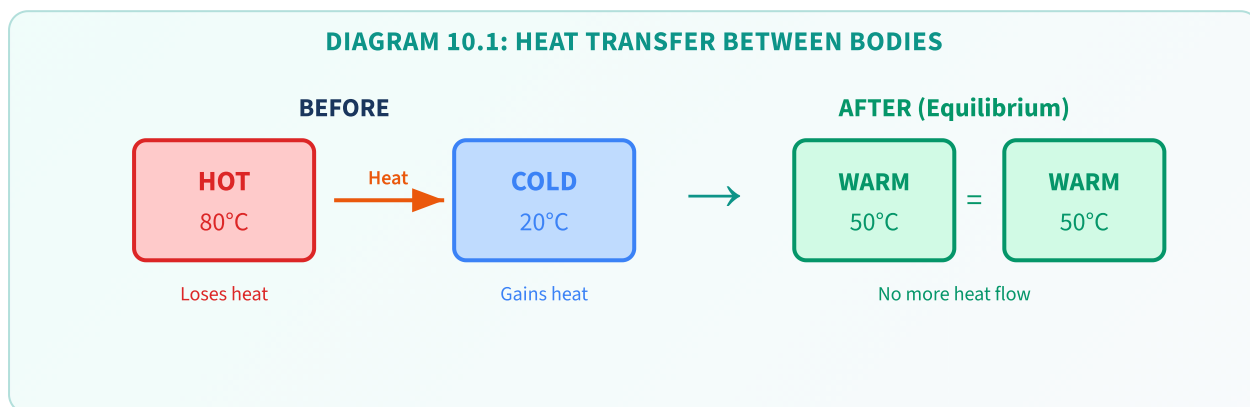
DEFINITION

Thermal Equilibrium: Two bodies are in thermal equilibrium when they are at the same temperature and no heat flows between them.

Heat Transfer Concept

When two bodies at different temperatures are brought in contact:

- Heat flows from hotter body (higher temperature) to cooler body (lower temperature)
- The hotter body loses heat and cools down
- The cooler body gains heat and warms up
- Transfer continues until both reach the same temperature (thermal equilibrium)



$$Q = mc\Delta T$$

Q = heat energy (J), m = mass (kg), c = specific heat capacity (J/kg·K)
 ΔT = change in temperature (K or °C)

10.3 Measurement of Temperature

Thermometers

DEFINITION

Thermometer: A device used to measure temperature. It works on the principle that some physical property of a substance changes uniformly with temperature (e.g., length of mercury column, resistance of wire, volume of gas).

Type of Thermometer	Property Used	Range
Mercury thermometer	Expansion of mercury	−39°C to 357°C
Alcohol thermometer	Expansion of alcohol	−115°C to 78°C
Constant volume gas thermometer	Pressure of gas at constant volume	Very wide range
Platinum resistance thermometer	Electrical resistance	−200°C to 1200°C
Thermocouple	EMF (Seebeck effect)	−200°C to 1600°C
Pyrometer	Radiation intensity	Above 800°C

Temperature Scales

Scale	Lower Fixed Point (Ice)	Upper Fixed Point (Steam)	Divisions
Celsius (°C)	0°C	100°C	100
Fahrenheit (°F)	32°F	212°F	180
Kelvin (K)	273.15 K	373.15 K	100

Conversion Formulas

$$\begin{aligned}\text{°F} &= (9/5)\text{°C} + 32 \\ \text{°C} &= (5/9)(\text{°F} - 32) \\ \text{K} &= \text{°C} + 273.15\end{aligned}$$

UNIVERSAL CONVERSION FORMULA

All scales are related by:

$$(\text{C} - 0)/100 = (\text{F} - 32)/180 = (\text{K} - 273)/100$$

Special point: $-40^{\circ}\text{C} = -40^{\circ}\text{F}$ (the two scales meet at this value)

EXAMPLE

Convert 37°C (human body temperature) to Fahrenheit and Kelvin.

$$\text{°F} = (9/5)(37) + 32 = 66.6 + 32 = \mathbf{98.6^{\circ}\text{F}}$$

$$\text{K} = 37 + 273.15 = \mathbf{310.15\text{ K}}$$

Key Points

- ✓ Kelvin is the SI unit of temperature (absolute scale, no negative values)
- ✓ $0\text{ K} = -273.15^{\circ}\text{C}$ = absolute zero (lowest possible temperature)
- ✓ Size of 1°C = size of 1 K (but different starting points)
- ✓ Size of $1^{\circ}\text{F} = 5/9$ of 1°C

10.4 Ideal Gas Equation

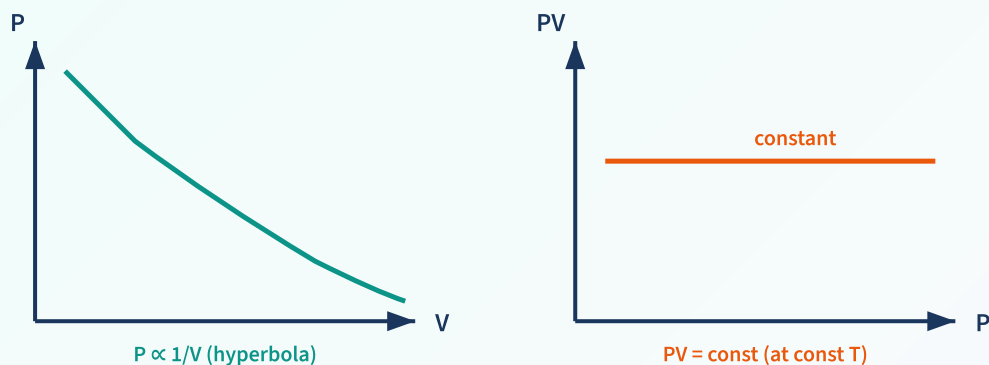
Boyle's Law (at constant temperature)

STATEMENT

Boyle's Law: At constant temperature, the volume of a given mass of gas is inversely proportional to its pressure.

$$\begin{aligned}\text{PV} &= \text{constant} \quad (\text{at constant } T) \\ P_1V_1 &= P_2V_2\end{aligned}$$

DIAGRAM 10.2: BOYLE'S LAW



Charles' Law (at constant pressure)

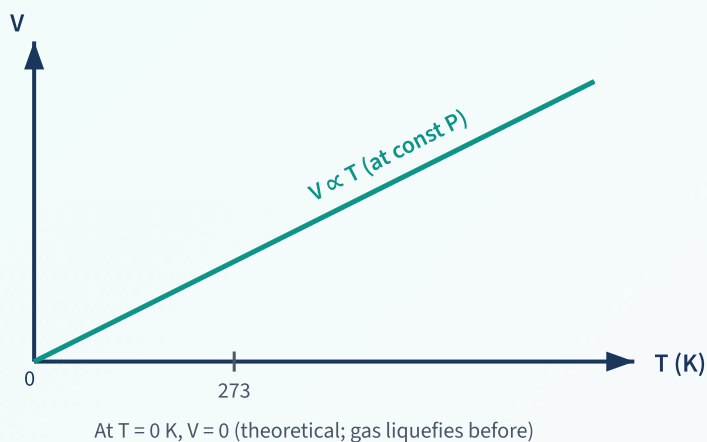
STATEMENT

Charles' Law: At constant pressure, the volume of a given mass of gas is directly proportional to its absolute temperature (in Kelvin).

$$V/T = \text{constant} \quad (\text{at constant } P)$$

$$V_1/T_1 = V_2/T_2$$

DIAGRAM 10.3: CHARLES' LAW



Ideal Gas Law

Combining Boyle's Law and Charles' Law:

$$PV = nRT$$

P = pressure (Pa), V = volume (m^3), n = number of moles

R = universal gas constant = **$8.314 \text{ J}/(\text{mol} \cdot \text{K})$**

T = absolute temperature (K)

$$PV = Nk_B T$$

N = total number of molecules, k_B = Boltzmann constant = 1.38×10^{-23} J/K

$R = N_a \times k_B$ (where N_a = Avogadro's number = 6.022×10^{23})

EXAMPLE

Find the volume of 2 moles of an ideal gas at STP ($T = 273$ K, $P = 1.013 \times 10^5$ Pa).

$$V = nRT/P = 2 \times 8.314 \times 273 / (1.013 \times 10^5)$$

$$V = 4539.4 / 101300 = \mathbf{0.0448 \text{ m}^3 \approx 44.8 \text{ litres}}$$

(1 mole at STP occupies 22.4 L)

Key Points

- ✓ Boyle's Law: $PV = \text{const}$ (at constant T); $P \propto 1/V$
- ✓ Charles' Law: $V/T = \text{const}$ (at constant P); $V \propto T$
- ✓ Gay-Lussac's Law: $P/T = \text{const}$ (at constant V); $P \propto T$
- ✓ Ideal Gas: $PV = nRT$; $R = 8.314$ J/(mol·K)
- ✓ At STP: 1 mole of any ideal gas occupies 22.4 litres

10.5 Thermal Expansion

Most substances expand when heated. This happens because molecules vibrate more vigorously at higher temperatures, increasing the average distance between them.

(a) Linear Expansion (Length)

DEFINITION

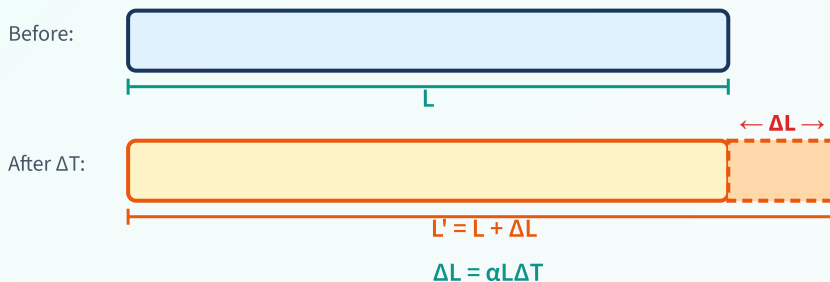
Linear Expansion: The increase in length of a solid rod when its temperature is raised. Applicable to one-dimensional objects (rods, wires).

$$\Delta L = \alpha L \Delta T \quad \rightarrow \quad L' = L(1 + \alpha \Delta T)$$

α = coefficient of linear expansion (unit: $/^{\circ}\text{C}$ or $/\text{K}$)

L = original length, ΔT = temperature change, L' = new length

DIAGRAM 10.4: LINEAR EXPANSION



(b) Area (Superficial) Expansion

$$\Delta A = \beta A \Delta T \rightarrow A' = A(1 + \beta \Delta T)$$

β = coefficient of area expansion

$$\beta = 2\alpha \text{ (relation between area and linear coefficients)}$$

(c) Volume (Cubical) Expansion

$$\Delta V = \gamma V \Delta T \rightarrow V' = V(1 + \gamma \Delta T)$$

γ = coefficient of volume expansion

$$\gamma = 3\alpha \text{ (relation between volume and linear coefficients)}$$

IMPORTANT RELATION

$$\alpha : \beta : \gamma = 1 : 2 : 3$$

This relation holds for isotropic solids (same properties in all directions). For liquids, only volume expansion (γ) is meaningful since they have no fixed shape.

Material	$\alpha (\times 10^{-6} / ^\circ\text{C})$	Use/Note
Steel	12	Railway tracks, bridges
Aluminum	23	Cooking vessels
Copper	17	Electrical wires
Glass	9	Pyrex has low α (3.2)
Invar (Ni-Fe alloy)	0.9	Clock pendulums, measuring tapes

EXAMPLE

A steel rail is 10 m long at 20°C. Find its length at 50°C. ($\alpha = 12 \times 10^{-6} / ^\circ\text{C}$)

$$\Delta L = \alpha L \Delta T = 12 \times 10^{-6} \times 10 \times 30 = 3.6 \times 10^{-3} \text{ m} = \mathbf{3.6 \text{ mm}}$$

$$L' = 10 + 0.0036 = \mathbf{10.0036 \text{ m}}$$

This is why gaps are left between railway tracks!

Key Points

- ✓ Linear: $\Delta L = \alpha L \Delta T$; Area: $\Delta A = \beta A \Delta T$; Volume: $\Delta V = \gamma V \Delta T$
- ✓ $\alpha : \beta : \gamma = 1 : 2 : 3$
- ✓ Water anomaly: water contracts from 0°C to 4°C (max density at 4°C)
- ✓ Gaps in railway tracks, expansion joints in bridges — allow for expansion
- ✓ Bimetallic strip: two metals with different α bonded together; bends on heating

10.6 Specific Heat Capacity

DEFINITION

Specific Heat Capacity (c): The amount of heat required to raise the temperature of unit mass (1 kg) of a substance by 1°C (or 1 K).

$$Q = mc\Delta T$$

Q = heat absorbed or released (J)

m = mass of substance (kg)

c = specific heat capacity (J/kg·K)

ΔT = change in temperature (K or °C)

UNDERSTANDING SPECIFIC HEAT

A substance with **high specific heat** (like water, $c = 4186 \text{ J/kg}\cdot\text{K}$) needs a lot of heat to raise its temperature — it heats up slowly and cools down slowly.

A substance with **low specific heat** (like iron, $c = 450 \text{ J/kg}\cdot\text{K}$) needs less heat — it heats up quickly and cools down quickly.

This is why water is used as a coolant, and why coastal areas have moderate climate (water absorbs/releases heat slowly).

Substance	$c \text{ (J/kg}\cdot\text{K)}$	Note
Water	4186	Highest among common substances
Ice	2100	About half of water
Steam	2010	About half of water
Aluminum	900	Good for cooking vessels
Iron/Steel	450	Heats quickly
Copper	390	Used in calorimeters
Lead	130	Very low — heats very fast

Molar Heat Capacity

$$C = Mc \quad (C = \text{molar heat capacity, } M = \text{molar mass})$$

Unit: J/(mol·K)

For gases: C_p (at constant pressure) and C_v (at constant volume)

$$C_p - C_v = R \text{ (Mayer's relation)}$$

EXAMPLE

How much heat is needed to raise temperature of 2 kg water from 20°C to 100°C?

$$Q = mc\Delta T = 2 \times 4186 \times (100 - 20) = 2 \times 4186 \times 80$$

$$Q = 669,760 \text{ J} \approx 670 \text{ kJ}$$

Key Points

- ✓ $Q = mc\Delta T$ (fundamental formula for heat exchange)
- ✓ Water has highest specific heat — excellent coolant
- ✓ Unit of c : J/(kg·K); higher $c \rightarrow$ slower heating/cooling
- ✓ For gases: $C_p > C_v$; $C_p - C_v = R$

10.7 Calorimetry

DEFINITION

Calorimetry: The measurement of heat exchange between bodies. A calorimeter is an insulated vessel used for this purpose.

Principle of Calorimetry (Heat Exchange)

PRINCIPLE

In an isolated system (no heat loss to surroundings), the heat lost by the hot body equals the heat gained by the cold body.

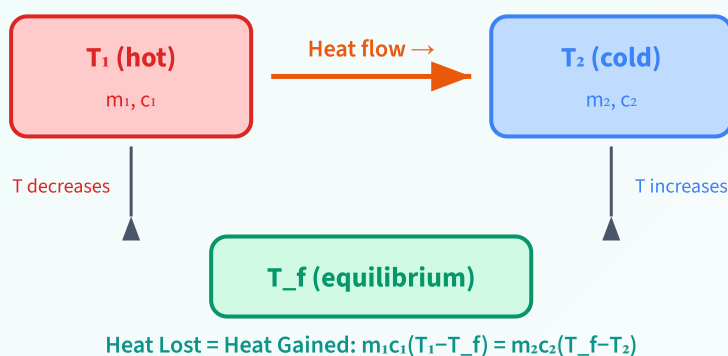
Heat Lost = Heat Gained

$$m_1 c_1 (T_1 - T_f) = m_2 c_2 (T_f - T_2)$$

T_1 = initial temperature of hot body, T_2 = initial temperature of cold body

T_f = final equilibrium temperature

DIAGRAM 10.5: CALORIMETRY



EXAMPLE

A 200 g iron ball at 100°C is dropped into 500 g water at 20°C. Find final temperature. ($c_{\text{iron}} = 450 \text{ J/kg}\cdot\text{K}$, $c_{\text{water}} = 4186 \text{ J/kg}\cdot\text{K}$)

Heat lost by iron = Heat gained by water

$$0.2 \times 450 \times (100 - T_f) = 0.5 \times 4186 \times (T_f - 20)$$

$$90(100 - T_f) = 2093(T_f - 20)$$

$$9000 - 90T_f = 2093T_f - 41860$$

$$50860 = 2183T_f$$

$$T_f = \mathbf{23.3^\circ\text{C}}$$

(Iron cools a lot, water barely warms — because water has much higher heat capacity!)

Key Points

- ✓ Principle: Heat lost = Heat gained (in isolated system)
- ✓ Final temperature lies between T_1 and T_2
- ✓ Calorimeter is usually made of copper (known c , good conductor)
- ✓ Include heat absorbed by calorimeter in calculations for accuracy

10.8 Change of State

DEFINITION

Change of State: The transition of matter from one physical state (solid, liquid, gas) to another. During change of state, temperature remains constant even though heat is being supplied or removed.

Latent Heat

DEFINITION

Latent Heat (L): The amount of heat required to change the state of unit mass of a substance without any change in temperature.

$$Q = mL$$

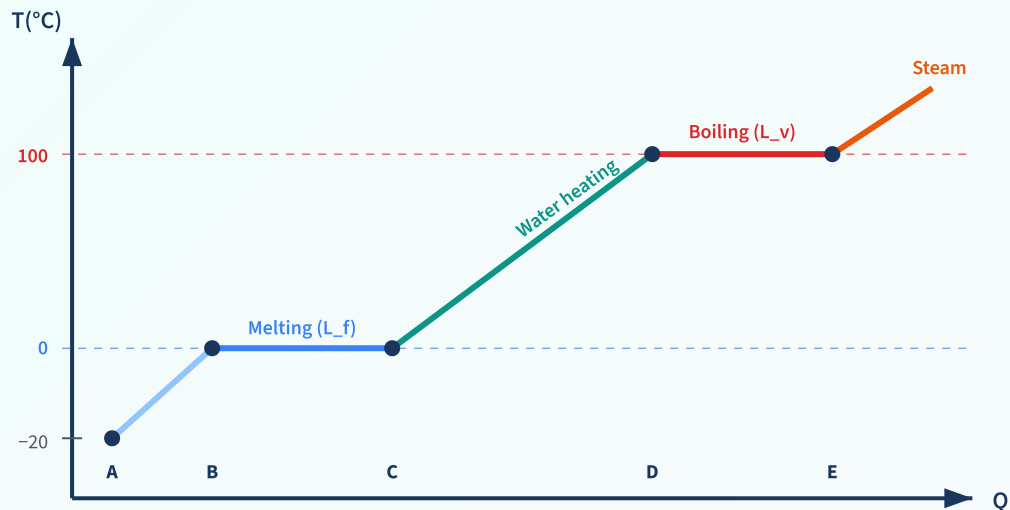
Q = heat (J), m = mass (kg), L = latent heat (J/kg)

No temperature change during phase transition!

Melting (Fusion) & Boiling (Vaporization)

Process	Transition	Latent Heat (for water)	Temperature
Melting (Fusion)	Solid $\rightarrow$ Liquid	$L_f = 3.34 \times 10^5$ J/kg	0°C (for ice)
Freezing	Liquid $\rightarrow$ Solid	Same L_f (released)	0°C
Boiling (Vaporization)	Liquid $\rightarrow$ Gas	$L_v = 22.6 \times 10^5$ J/kg	100°C (for water)
Condensation	Gas $\rightarrow$ Liquid	Same L_v (released)	100°C
Sublimation	Solid $\rightarrow$ Gas (directly)	$L_f + L_v$	Varies

DIAGRAM 10.6: HEATING CURVE OF WATER (ICE TO STEAM)



A→B: Ice heats B→C: Ice melts at 0°C ($Q = mL_f$) C→D: Water heats D→E: Boils at 100°C ($Q = mL_v$) E→∞: Steam heats

EXAMPLE

Find heat needed to convert 1 kg of ice at -10°C to steam at 100°C .

Step 1: Heat ice from -10°C to 0°C : $Q_1 = mc_{\text{ice}}\Delta T = 1 \times 2100 \times 10 = 21,000 \text{ J}$

Step 2: Melt ice at 0°C : $Q_2 = mL_f = 1 \times 3.34 \times 10^5 = 3,34,000 \text{ J}$

Step 3: Heat water from 0°C to 100°C : $Q_3 = mc_w\Delta T = 1 \times 4186 \times 100 = 4,18,600 \text{ J}$

Step 4: Boil water at 100°C : $Q_4 = mL_v = 1 \times 22.6 \times 10^5 = 22,60,000 \text{ J}$

Total: $Q = 21,000 + 3,34,000 + 4,18,600 + 22,60,000 = 30,33,600 \text{ J} \approx 3034 \text{ kJ}$

Key Points

- ✓ $Q = mL$ (heat for change of state, no temperature change)
- ✓ L_f (ice) = $3.34 \times 10^5 \text{ J/kg}$; L_v (water) = $22.6 \times 10^5 \text{ J/kg}$
- ✓ $L_v \gg L_f$ (boiling needs much more energy than melting)
- ✓ Steam burns are worse than water burns (extra latent heat released)
- ✓ Pressure affects melting/boiling point (increase in $P \rightarrow$ higher BP, lower MP for water)

10.9 Heat Transfer

Heat can be transferred from one place to another by three methods:

(a) Conduction

DEFINITION

Conduction: Transfer of heat through a material without actual movement of the material itself. Heat passes from molecule to molecule by vibration. Occurs mainly in solids.

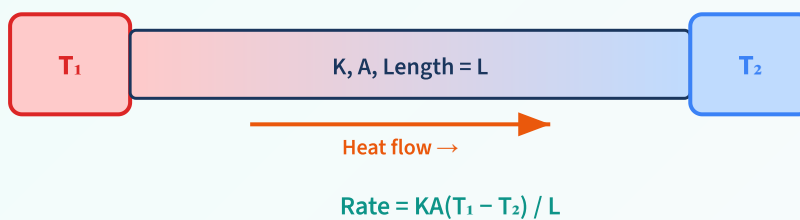
$$Q/t = KA(T_1 - T_2)/L$$

Q/t = rate of heat flow (W), K = thermal conductivity ($W/m \cdot K$)

A = cross-sectional area, L = length/thickness

$T_1 - T_2$ = temperature difference

DIAGRAM 10.7: CONDUCTION THROUGH A ROD



EXAMPLES

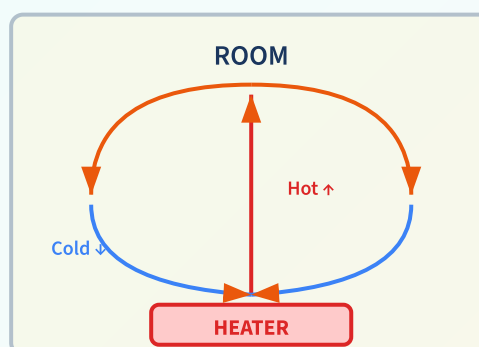
- Metal spoon becomes hot when dipped in hot soup (good conductor)
- Wooden handle stays cool (bad conductor/insulator)
- Winter clothes trap air (poor conductor) → keep you warm

(b) Convection

DEFINITION

Convection: Transfer of heat by actual movement (flow) of the heated fluid (liquid or gas). Hot fluid rises (less dense), cold fluid sinks (more dense), creating convection currents.

DIAGRAM 10.8: CONVECTION CURRENTS



Hot fluid rises → Cold fluid sinks → Circular convection

EXAMPLES

- **Sea breeze/Land breeze:** Convection due to unequal heating of land and sea
- **Boiling water:** Hot water rises from bottom, cold water sinks
- **Room heater:** Placed at bottom; warm air rises and circulates
- **Ventilation:** Hot air exits from top, fresh cool air enters from below

(c) Radiation

DEFINITION

Radiation: Transfer of heat in the form of electromagnetic waves. It does NOT require any medium — heat can travel through vacuum (e.g., sunlight reaching Earth).

PROPERTIES OF THERMAL RADIATION

- Travels at the speed of light (3×10^8 m/s)
- Does not need a medium (travels through vacuum)
- Travels in straight lines
- Can be reflected and absorbed like light
- Dark/rough surfaces absorb and emit more radiation than shiny/smooth surfaces

Stefan-Boltzmann Law

$$E = \sigma AT^4$$

E = energy radiated per second (W), σ = Stefan's constant = 5.67×10^{-8} W/m²K⁴

A = surface area, T = absolute temperature (K)

For a body at T in surroundings at T_0 : Net radiation = $\sigma A(T^4 - T_0^4)$

BLACK BODY

A **perfect black body** absorbs all radiation falling on it and also emits maximum radiation at any temperature. It is an ideal concept — lamp black and platinum black are close approximations. Wien's Law: $\lambda_{\text{max}} \times T = \text{constant}$ (hotter body → shorter peak wavelength).

Property	Conduction	Convection	Radiation
Medium needed?	Yes (solids mainly)	Yes (fluids only)	No (works in vacuum)
Material movement?	No	Yes (bulk flow)	No
Speed	Slow	Moderate	Speed of light
Direction	Along temperature gradient	Vertical (up/down)	All directions (straight lines)
Example	Hot pan handle	Boiling water	Sunlight, campfire warmth

📌 Key Points

- ✓ Conduction: molecule-to-molecule (solids); Rate = $KA(T_1 - T_2)/L$
- ✓ Convection: actual fluid movement (liquids/gases); hot rises, cold sinks
- ✓ Radiation: EM waves, no medium needed; $E = \sigma AT^4$
- ✓ Black body: perfect absorber and emitter of radiation
- ✓ Dark surfaces absorb/emit more; shiny surfaces reflect more

10.10 Newton's Law of Cooling

STATEMENT

Newton's Law of Cooling: The rate of loss of heat (or rate of cooling) of a body is directly proportional to the difference in temperature between the body and its surroundings, provided the temperature difference is small.

$$dT/dt = -k(T - T_0)$$

T = temperature of body at time t

T_0 = temperature of surroundings (constant)

k = cooling constant (depends on surface area, nature of surface)

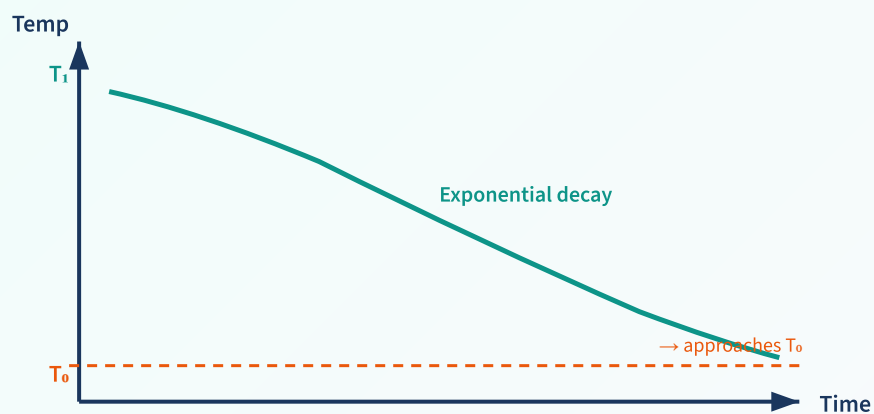
Negative sign: temperature decreases with time

$$\text{Approximate form: } (T_1 - T_2)/t = k[(T_1 + T_2)/2 - T_0]$$

T_1 = initial temp, T_2 = final temp, t = time taken

Useful for numerical problems

DIAGRAM 10.9: NEWTON'S LAW OF COOLING — GRAPH



Body cools fast initially, rate decreases as it approaches surroundings temp

EXAMPLE

A body cools from 80°C to 70°C in 5 minutes. Room temperature is 25°C. Find time to cool from 70°C to 60°C.

Case 1: $(80-70)/5 = k[(80+70)/2 - 25] \rightarrow 2 = k(75-25) = 50k \rightarrow k = 0.04$

Case 2: $(70-60)/t = 0.04[(70+60)/2 - 25] \rightarrow 10/t = 0.04(65-25) = 0.04 \times 40 = 1.6$

$t = 10/1.6 = \mathbf{6.25 \text{ minutes}}$

(Takes longer because temperature difference with surroundings is smaller)

Key Points

- ✓ Rate of cooling $\propto (T - T_0)$ — valid for small temperature differences
- ✓ Cooling is faster when temperature difference is large
- ✓ Graph: exponential decay approaching T_0 asymptotically
- ✓ Hot tea cools fast initially, then slower and slower
- ✓ Approximate formula useful for problems: $(T_1 - T_2)/t = k[(T_1 + T_2)/2 - T_0]$

Quick Revision Summary - Chapter 10

- ✓ **Heat:** Energy transfer due to temperature difference; Unit: Joule
- ✓ **Temperature:** Degree of hotness; Unit: Kelvin; $^{\circ}\text{F} = (9/5)^{\circ}\text{C} + 32$; $\text{K} = ^{\circ}\text{C} + 273$
- ✓ **Ideal Gas:** $PV = nRT$; Boyle: $PV = \text{const}$; Charles: $V/T = \text{const}$
- ✓ **Linear expansion:** $\Delta L = \alpha L \Delta T$; $\alpha : \beta : \gamma = 1 : 2 : 3$
- ✓ **Specific heat:** $Q = mc\Delta T$; water has highest c (4186 J/kg·K)
- ✓ **Calorimetry:** Heat lost = Heat gained
- ✓ **Latent heat:** $Q = mL$; $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J/kg}$; $L_v(\text{water}) = 22.6 \times 10^5 \text{ J/kg}$
- ✓ **Conduction:** $Q/t = KA(T_1 - T_2)/L$ (through solids)
- ✓ **Convection:** Bulk fluid movement; hot rises, cold sinks
- ✓ **Radiation:** EM waves; no medium; $E = \sigma AT^4$
- ✓ **Newton's cooling:** $dT/dt = -k(T - T_0)$; exponential decay

Practice Questions

Very Short Answer Questions (1 Mark)

1. Define specific heat capacity.
2. What is latent heat?
3. Convert 100°F to Celsius.
4. State Newton's Law of Cooling.
5. What is thermal conductivity?
6. Why is water used as a coolant in car engines?
7. What is the relation between α , β , and γ ?

8. Define absolute zero.
9. Why does steam at 100°C cause more severe burns than water at 100°C ?
10. State Boyle's Law.

Short Answer Questions (2-3 Marks)

1. Distinguish between heat and temperature with examples.
2. Explain linear, area, and volume expansion. How are their coefficients related?
3. State and explain the principle of calorimetry.
4. What is anomalous expansion of water? Why is it important for aquatic life?
5. Explain the difference between conduction, convection, and radiation.
6. Why are gaps left between railway tracks?
7. Derive the ideal gas equation $PV = nRT$ from Boyle's and Charles' Laws.
8. Explain why coastal areas have moderate climate using specific heat concept.
9. Draw and explain the heating curve of ice to steam.
10. What is a black body? State Stefan-Boltzmann law.

Long Answer Questions (5 Marks)

1. Define thermal conductivity. Derive the expression for rate of heat flow through a rod. What factors affect it?
2. Explain the three modes of heat transfer with examples, diagrams, and one formula for each.
3. State and derive Newton's Law of Cooling. Draw the cooling curve and explain its nature.
4. What is latent heat? Explain the complete process of converting ice at -20°C to steam at 120°C with all the heat calculations.
5. Explain thermal expansion in detail. Derive the relation between α , β , and γ . Give applications and examples of thermal expansion.

Numerical Problems

1. Convert: (a) 37°C to $^{\circ}\text{F}$ and K, (b) 98.6°F to $^{\circ}\text{C}$, (c) 400 K to $^{\circ}\text{C}$.
2. A steel bridge is 200 m long at 10°C . Find its length at 40°C . ($\alpha = 12 \times 10^{-6}/^{\circ}\text{C}$)
3. How much heat is needed to raise 5 kg of water from 25°C to 100°C ? ($c = 4186 \text{ J/kg}\cdot\text{K}$)
4. Find the heat required to convert 2 kg of ice at 0°C to water at 20°C . ($L_f = 3.34 \times 10^5 \text{ J/kg}$, $c_w = 4186 \text{ J/kg}\cdot\text{K}$)
5. A 500 g copper ball at 200°C is dropped in 2 kg water at 30°C . Find final temperature. ($c_{\text{Cu}} = 390$, $c_w = 4186 \text{ J/kg}\cdot\text{K}$)
6. A metal rod ($K = 50 \text{ W/m}\cdot\text{K}$, $A = 5 \text{ cm}^2$, $L = 1 \text{ m}$) has ends at 100°C and 20°C . Find rate of heat transfer.
7. 2 moles of an ideal gas at 300 K occupy 0.05 m^3 . Find the pressure.
8. A body cools from 60°C to 50°C in 10 min. Room temp = 25°C . Find time to cool from 50°C to 40°C .
9. Find heat needed to convert 500 g ice at -10°C to steam at 100°C . ($c_{\text{ice}} = 2100$, $L_f = 3.34 \times 10^5$, $c_w = 4186$, $L_v = 22.6 \times 10^5$)
10. A glass flask (volume 250 cm^3) is filled with mercury at 20°C . How much mercury overflows when heated to 100°C ? ($\gamma_{\text{glass}} = 27 \times 10^{-6}/^{\circ}\text{C}$, $\gamma_{\text{mercury}} = 182 \times 10^{-6}/^{\circ}\text{C}$)

Application-Based Questions

1. Why do we wear light-coloured clothes in summer and dark-coloured in winter?
2. Explain how a thermos flask minimizes all three modes of heat transfer.
3. Why is frost more likely on a clear night than a cloudy night?
4. Why does a metal chair feel colder than a wooden chair at the same temperature?
5. Explain how sea breeze and land breeze work using convection.